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Sustainable Transition and Business Model Development

A Case Study on the Swedish Forest Industry

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Abstract

Sustainable transitions in industries are increasingly important. The Swedish forest industry is making efforts by creating the project TREE to help decarbonize their transport sector, gathering actors throughout their industry. It serves as a good basis for exploring the challenges that occur when transitioning in an ecosystem, both in terms of their internal difficulties, but also their alignment when contributing to the transition in which they are dependent on each other. The aim was to explore these challenges that occur from an organizational- and ecosystem perspective related to business model development. The three literature streams of transition theory, innovation ecosystems and business model development formed a foundation where both the aspects of organizations and ecosystems were explored. Following that, an exploratory study was conducted with data gathered mainly from qualitative interviews with people working in- and around the defined ecosystem of actors within the TREE project. A total of 13 interviews were made with 12 informants, of which three were considered external to the defined ecosystem. The data was analyzed with an open and explorative approach inspired by Gioia (Corley and Gioia, 2004) to uncover new insights, which resulted in five aggregated dimensions (*Technological barriers, Project serves different purposes, Haulers dependent on forest company, Resistance to change and Decision-making shifting inside transport companies*) representing the main findings. The results gave a number of prominent challenges both regarding business model development and that of collective value. Internal tensions and the changing role of the driver were the most prominent challenges of business model development. For ecosystems, capturing the collective value and running multiple business models stood out as the greatest challenges. Digitalization emerged as a requirement for the whole ecosystem to transition, and the role of transport provider was discussed in relation to a previous study. Lastly, two tensions and two paradoxes were identified, which were *Control vs. Flexibility*, *Project vs. Business* as tensions and *Competition vs. Collaboration*, *Dependency vs. Change* as the paradoxes. The master's thesis contributes with valuable considerations for sustainable transitions with similar ecosystem setups, or in an equally early stage.

Keywords

Business model development, Sustainable transition, Innovation ecosystem, Electrification, Forest industry

Sammanfattning

Hållbara omställningar inom industrin blir allt viktigare. Den svenska skogsindustrin tar steg i denna riktning genom projektet TREE, som syftar till att minska koldioxidutsläppen inom transportsektorn genom att samla aktörer från hela branschen. Projektet utgör en god grund för att undersöka de utmaningar som uppstår vid en omställning inom ett ekosystem – både utifrån interna svårigheter och den samordning som krävs när aktörerna är ömsesidigt beroende i omställningen. Syftet med studien var att utforska dessa utmaningar ur ett organisatoriskt och ekosystemperspektiv, med särskilt fokus på affärsmodellsutveckling. Tre litteraturströmmar – transitionsteori, innovationsekosystem och affärsmodellsutveckling – låg till grund för att undersöka både organisatoriska och ekosystemrelaterade aspekter. Studien genomfördes som en explorativ fallstudie med datainsamling huvudsakligen genom kvalitativa intervjuer med personer verksamma inom och kring det definierade ekosystemet av aktörer i TREE-projektet. Totalt genomfördes 13 intervjuer med 12 informanter, varav tre ansågs vara externa i förhållande till det definierade ekosystemet. Data analyserades med en öppen och utforskande metod inspirerad av Gioia (Corley och Gioia, 2004), vilket möjliggjorde upptäckten av nya insikter. Resultatet av analysen resulterade i fem aggregerade dimensioner (*Teknologiska hinder, Projektet tjänar olika syften, Åkerier är beroende av skogsbolagen, Motstånd mot förändring samt Beslutsfattande förskjuts inom transportföretagen*) som representerar studiens huvudsakliga resultat. Resultaten visade på flera framträdande utmaningar, både vad gäller affärsmodellsutveckling och kollektiva värdeskapandeprocesser. Interna spänningar samt förarens förändrade roll utgjorde de största utmaningarna kopplade till affärsmodeller, medan värdefångst i ekosystemet och parallella affärsmodeller framstod som särskilt utmanande för ekosystemet i stort. Digitalisering framkom som en nödvändighet för hela ekosystemets omställning, och transportörens roll diskuterades i relation till tidigare forskning. Slutligen identifierades två spänningar och två paradoxer: *Kontroll vs. Flexibilitet* och *Projekt vs. Affär* som spänningar, samt *Konkurrens vs. Samarbete* och *Beroende vs. Förändringskraft* som paradoxer. Detta examensarbete bidrar med värdefulla insikter för hållbara omställningar i liknande ekosystemstrukturer, särskilt i ett tidigt skede av omställningen.

Nyckelord

Affärsmodellsutveckling, Hållbar omställning, Innovationsekosystem, Elektrifiering, Skogsindustri

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Chapter 1

Introduction

The following chapter provides the background of the research, its purpose and its delimitation.

1.1 Background

The transition towards sustainable practices has become essential across industries in response to climate change and increasing regulatory and societal pressure. Technological advancements provide new opportunities for lowering the environmental impact of organisations. Yet, their adoption still remains to be a significant challenge. Transitioning to new technologies is complex and involves often more effort than just replacing technology itself.

Among these industries, the transport sector, particularly heavy-duty road transportation, remains one of the most challenging sectors to decarbonize. In recent years battery electric vehicles (BEV) have emerged as a promising solution. While progress has been made in passenger vehicles, the transition to BEV trucks for heavy road transport is still facing numerous technical, economic, and operational challenges.

Within this context, the Swedish forest industry has begun to investigate sustainable alternatives to a more sustainable road transportation. The Swedish forest industry is a substantial part of the national economy, accountable for nine to 12% of total jobs, exports, turnover and value added (Swedish Forest Industries, 2024). The transport sector is essential to the industry's supply chain. It connects remote harvesting areas with processing facilities and export terminals. As a result, the forest industry is Sweden's biggest transport buyer, accountable for 20% of heavy road transportation nationwide, which results in a significant contribution to the CO2 emissions

on Swedish roads (Skogforsk.se, 2023).

To address this issue, the TREE (Transition to Efficient Electrified forestry transportation) project brings together actors from across the Swedish forest industry's value chain to collaboratively accelerate the transition towards a more sustainable future, with a specific focus on electrifying heavy road transports. The project is in an early stage and has by the start of this research only a few trucks running in operation. It is divided into seven different demonstration sites across Sweden, each with its unique features and challenges. These sites include different combinations of actors such as forest companies, transport companies, haulers and charging providers, making it a complex multi-business environment. The overall goal of this initiative is to have 50% of the newly introduced heavy-duty truck fleet supplying the Swedish forest industry electrified by 2030. This ambitious target, however, requires overcoming strategic, technological, and systemic barriers that are deeply embedded in the operations and relationships of the actors involved.

1.2 Purpose

Sustainable transitions require new business models in order to be operationally and economically viable. Stakeholders within the transitioning industry create a network of actors driving for the same goal. What can emerge in a successful innovation ecosystem, can also bring challenges due to each firm's individual needs. This can raise tensions in terms of value creation, capture and delivery. It is important to consider how the actors are sharing the risk and handle uncertainty both individually and collectively. This requires coordination between the actors and strategies for addressing the different challenges that occur.

The challenges can be different depending on the perspective and needs a holistic approach to properly identify the nature of them. The TREE project brings such challenges, where the forestry companies are the main drivers of these transition efforts, which provides certain circumstances that impact the transition. An important consideration of a successful transition in this project is the complex interrelation between the different actors in this innovation ecosystem. Just as it is important for the individual actors to develop their business models in the changing environment, it is even more important that these developments contribute to the positive transition of the whole ecosystem. The aim of this master thesis is to explore the challenges regarding business model development that arise when accelerating the sustainable transition of an industry, both on an organizational level and

how that impacts the ecosystem.

1.3 Delimitations

Due to the extensiveness of this project and the limited timeframe available for a masters thesis, this research focuses purely on one specific site within the TREE project. Since this is a master thesis, a certain amount of coherence must be kept to the subjects of the respective institution the authors are students of. Detailed technical assessments of vehicle performance, cost-benefit analysis, total cost analysis, total cost of ownership and policy evaluations are beyond the scope of this thesis.

Chapter 2

Theoretical Framework

The aim of this master thesis is to explore the challenges regarding business model development that arise when accelerating the sustainable transition of an industry, both on an organizational level and how that impacts the ecosystem. The transition can have implications on the business model development for individual firms participating in the transition, but also the ecosystem as a whole. There are a few literature streams that study this phenomenon. Transition theory investigates the emergence of a new technology on different levels, which is important in the study of industries undergoing a sustainable transition and the establishment of new technologies. Business model innovation/development considers the individual organization and their survival in the changing environment. Innovation ecosystems somewhat ties them together in explaining how different individual firms interact in a system to contribute (or hinder) an innovation. Below follows a more detailed study of each literature stream to investigate their connections to this topic further and see the connections between them.

2.1 Transition Theory

Transitions in industries are almost always due to a change in- or emergence of a novel technology (Foster, 1985). In recent years, transitions to more sustainable practices has also gained more attention. There are numerous uncertainties and barriers appearing in the transition to low-carbon systems. Price, performance and social acceptance are a few examples. The current technologies and solutions are well-established to such an extent that their solution-space is limiting that of the emerging one through cognitive bias, which makes it difficult for the transition to follow through (Geels et al., 2017).

At the same time, technological changes have a tendency to make individual firms fall behind and miss the opportunity of the transition, which makes them irrelevant (Tongur and Engwall, 2014). To be part of the transition, individual organizations must both contribute and see to their own survival.

Industry transitions can be viewed from a multi-level perspective, with the niche regime, socio-technical regime and the socio-technical landscape as the general levels. The innovation typically starts in a niche market, where it grows and succeeds before scaling. Sometime during the transition, there could be a state of hybridization when the old and the new technology complement each other and harmonize. User-practices often change when transitioning to a new technology. A technological transition can impact all levels depending on the extensiveness of the innovation. Transport systems have previously been associated with breakthroughs at the socio-technical regime, since support functions and the infrastructure around also change to facilitate the new technology (Geels, 2002). Geels (2002) shows this through the transition from sailing ships to steam-ships, which radically changed the infrastructure with the mining and distribution of coal, as well as iron mining for building the actual ships. The analogy to electric timber trucks is not far away. Just like iron steamships, the electric truck is a radically different way of building trucks compared to those powered by combustion engines. Coal mining and distribution can be compared to the greater need for electricity and charging stations.

Some scholars have put sustainability into the context of transition theory. As there is a push for greater sustainability efforts and to lower the dependence on carbon-intensive technologies, which require whole socio-technical regimes to transition, sustainable transitions have become a popular research topic. Just like other technological transitions, it has long-term and fundamentally changes the socio-technical system. Unlike the traditional technological transition, the sustainable transition might happen because of what Köhler et al. (2019) describes as the flip-side of transitions, which is a deliberate way of changing systems. Policies and governments are likely the cause for the push and the use of long-term goals are there to guide the transition (Markard, Raven, and Truffer, 2012). Individual actors can also be the drivers of the transition, but will likely find themselves in a resistant environment (Farla et al., 2012). Sustainability transitions can take the form of incremental changes of the already established systems, but Markard, Raven, and Truffer (2012) deems this insufficient and calls for radical changes to facilitate a proper sustainable transition. Change is described from this perspective as long-term, dependent on several actors, dimensions and open-

ended (Köhler et al., 2019). Just as individual firms have to respond to their environments to stay competitive (Teece, 2018), some of the firms in the socio-technical system will lose value when addressing sustainability (Köhler et al., 2019). Firms that act in these regimes are bound for a changing environment and need to navigate in the transition.

Transition theory explains how the technological transition shifts from a small to a large scale, and considers the success of the technology. For a technological transition to happen, these elements need to be in place. For an emerging technological transition, it is interesting to investigate to what extent they are in place and if the niche-market has the possibility to scale-up to the socio-technical regime. Transition theory through multi-level analysis is also just one way of viewing an industry in transition, and important factors to the transition are most likely left out. Therefore, theory of the individual organizations business model development needs to be studied as well to understand the challenges of transitions.

2.2 Business Model Development

Innovations like electric trucks and other sustainability practices are changing the market that firms are operating in. Sometimes they induce uncertainty which is very important for the individual firm to handle. The abilities to sense a change, seize the opportunity and transform the organization is what (Teece, 2018) calls dynamic capabilities. In a changing environment facing uncertainty like that of the TREE project, firms need to deploy these dynamic capabilities to help form the strategy and increase their chances of transforming successfully. This needs to be done continuously. Further, dynamic capabilities and forming strategy is tightly coupled with business model development. The business model is a way of appropriating the value of a new innovation, e.g. a new green technology (Teece, 2018). Business models are defined as the architecture from which a firm creates, delivers and captures value . The different building blocks of the business model each contribute to either creating, capturing or delivering value (Osterwalder, Pigneur, and Clark, 2010; Teece, 2018).

A new technology can enable the usage of an already existing business model. More commonly though, it needs new business models to be utilized well enough and capture value from it (Chesbrough and Rosenbloom, 2002). Still, newly developed business models for firms are typically closely related to their old business model (Teece, 2018). Initially, it can be run alongside

the old, established way of conducting business. In what manner the new business model should be incorporated is up to each firm. Some firms treat the new business as a totally separate entity, while others keep it close to the core business. Over time, the new venture will either be integrated or separated from the established business, depending on the synergies (Osterwalder, Pigneur, and Clark, 2010). The manner in which this is done thus dictates whether or not a firm can be transformed successfully (Teece, 2018) and needs to be done continuously to create new value offerings (Atkova et al., 2025). This can involve forming new partnerships in order to acquire new competencies (Osterwalder, Pigneur, and Clark, 2010).

Developing sustainable business models often comes with tensions between different actors. Their goals and strategies for sustainability are not seldom different and create tensions and paradoxes when building business models. Some examples of tensions are 'performing' (between stakeholders, internal and external) and 'organizing' (conflict with established ways of conducting business in an organization). The tensions can arise in an organization, between firms and on a systemic level (Van Bommel, 2018). There are different strategies for handling these tensions which can guide firms when developing/building new business models addressing sustainability (Stubbs, 2019; Van Bommel, 2018). Hahn et al. (2015) gives the example of electric cars and proposes a spatial separation strategy, where the sustainability effort runs in parallel with the normal business to avoid tensions. That and other strategies can be relevant to investigate for this transition.

For the individual firms, challenges can also arise in the value offering when building sustainable business models. First, customers can be sceptical of the new offering, which reduces trust in the company. Secondly, it can also trigger price inflation from the use of expensive materials. Atkova et al. (2025) proposes customer engagement and communication to overcome this, and to advertise the added value the new offer brings. Value is expected to flow to these new business models, if their value is important for customers. Previous research shows that value can flow differently if the added value is incorporated in the primary business model or run as a secondary, and that value often migrates within the same industry (Hacklin, Björkdahl, and Wallin, 2018), which would be important to consider in this transition. In the light of system transitions and the importance of whole industries to transition, created value can also be viewed from a collective level (Oskam, Bossink, and Man, 2021).

Although value offering is largely the individual organization's entity,

most business models depend on other organizations in the same business area and partners. How value is appropriated is also dependent on the development and technological transition. Transition theory describes the transition of the technology and implications for the different regime-levels, but does not explain how or why organizations interact with each other in the emergence of a new technology to form these regimes. Similarly, business model development explains the interaction with other actors only from the individual organization's perspective. Thus, a better understanding of the forming of the so-called regimes and the role of individual firms in creating those regimes is needed.

2.3 Innovation Ecosystem

In the sustainable transition of an industry, different actors often combine efforts to strive for the same goal; to reduce the environmental impact of the industry. Therefore, these actors create an inter-organizational network and form collaborative and competitive relationships, which are the characteristics of an innovation ecosystem as described by Adner (2016). Innovation ecosystems can be focused either around one organization and its value proposition, a new technology or a platform (Jacobides, Cennamo, and Gawer, 2018). Moore (1996) introduced the term ecosystem by describing the analogies of natural ecosystems with the mechanisms related to businesses and innovation. The actors within the project emerge naturally, as a result of collaborational needs and shared objectives Russell and Smorodinskaya (2018) which need alignment of activities to contribute to a collective value proposition. Therefore, ecosystems can have many different types of actors and structural compositions (Adner, 2016). Regarding innovation efforts a driving company plays an important role for an effective ecosystem. Adner (2016) argues that this actor has a high influence on the health of the ecosystem's members. Despite sometimes being labeled as big and incumbent companies, they can be the drivers of sustainable transitions (Köhler et al., 2019). For innovation ecosystems it is important to sense which venture brings added value to the project and how the value generation is distributed among the actors (Lubik and Garnsey, 2014), which has been stressed before by other scholars.

In innovation ecosystems, new actors might be introduced as part of the transition. New partnerships can be associated with additional risks, however, they are still a significant factor for growth since one company can't achieve it alone (Lubik, Garnsey, and Minshall, 2013; Lubik and Garnsey, 2014). On

the other hand Kim and Vonortas (2014) argue that knowledge from external sources can even reduce technological risk through insights and guidance from additional actors with their respective expertise. However, for the complex transition of introducing electric trucks in their current operation, many stakeholders are involved, which also brings many uncertainties. Vasconcelos Gomes (2018) describe in their article the importance of differentiating individual uncertainties and collective uncertainties to successfully coordinate a complex network of actors. Some companies face uncertainties regarding only their business, however, this can naturally spread across the whole innovation ecosystem due to their interrelation and turn into a collective uncertainty (Vasconcelos Gomes, 2018). Once again, the importance of the individual and collective is showcased.

Innovation ecosystems explains how firms interact with each other to form a system, highly dependent on the individual organizations, to form a collective value. In a transitioning industry, this is important to consider since it both indicates the successful transition of an individual firm, and also the transition of the ecosystem, which connects the topics of transition theory and business model development. Moreover, many phenomena seem to occur through innovation ecosystems that hinder the transition to a new technology. In what instances they do occur and what implications it has to the technological transition, can yet be explored even further and be put into context of both the individual organization and the socio-technical system.

2.4 Research Questions

As stated by previous research, the actions of both individual firms and the ecosystem as a whole is of the utmost importance for sustainable transitions to be successful and accelerate. Actors need to navigate their role in the developing business environment with regards to their own survival, but also to contribute to the greater picture. This requires delicate and difficult decisions in the development of strategy and business models for the individual actors, in order to stay relevant and to contribute to the transition. Despite having similar goals, the transition of the ecosystem will most likely reveal barriers and bottlenecks inside and between organizations as the transition progresses. This leads to the research questions to investigate what challenges are currently posed on the ecosystem- and organizational level for accelerating the sustainable transition:

- What are the most prominent challenges developing business models when accelerating the transition to electrified heavy road transports?

- What challenges occur creating collective value in an ecosystem when accelerating the transition to electrified heavy road transports?

Chapter 3

Methods

This research investigates the challenges with business model development and creating collective value when transitioning to electric vehicles in an industry. This phenomena was examined based on a case research setting of the TREE project, which aims to introduce electric trucks in the Swedish forest industry. The electrification of the forest industry's truck fleet is a novel approach to a green transition. Therefore, this research is of exploratory nature and aims to investigate the burdens which hinders the transition.

3.1 Empirical Setting

The empirical setting is a particular case within a research project. Below follows a more detailed description of the setting and its implications on the research.

3.1.1 TREE Project

The research was conducted within the TREE project. The project is coordinated by the research institute Skogforsk, which is a joint research institute by the forestry industry and together with the network organization CLOSER, they are based in Lindholmen. The project is funded by Vinnova within the framework of the RD programme FFI (Strategic Vehicle Research and Innovation) and is set to run 36 months and start in November 2023. Vinnova is an innovation agency, who works on behalf of the government. The total project budget is set to 157 million SEK, including 67 million SEK governmental funding. This project represents a complex system where many parameters are dependent on each other. The project involves 23 partners

across Sweden with the aim to cover all sectors to create an ecosystem. Partners include:

- Vehicle Manufacturer
- Forest Companies
- Logistic Companies and Haulers
- Technology Companies
- Universities

Due to the size of the project, it is divided into seven different sites to capture the variation between different parts of the country. These sites are utilized as hubs for knowledge building, involving research and development. Knowledge sharing is central to the project. Therefore, they develop activities and workshops to exchange experiences. Publications and results are updated regularly with reports. Currently, there are 12 electric timber and wood chip trucks operating within the project, including a 94 ton HCT vehicle. More efforts within the project setting are the investigations of electric long haul trailers, mobile energy hubs and the development and use of advanced route planning which are tested in real operations (Skogforsk.se, 2023).

3.1.2 Focus Area

The project is also divided into various work packages, one of which deals with business models. Within this work package the TREE project seeks support from researchers from KTH, which allowed the authors to work closely together with them. Due to delays, the timber truck started operations in the beginning of March 2025, and since then carefully followed and tested to capture insights. Site 4 has favourable conditions for the transition. It is located in a more urbanized area, which means the charging infrastructure is one of the most developed out of the different sites. It is split into two sub-sites, one of which is mainly operating wood chip trucks, while the other is primarily dependent on timber truck operations. The operation of those trucks are distinctly different. While the wood chip truck has its set routes, delivering between stationary delivering locations, the timber trucks routes are dynamic due to changing harvesting sites. This introduces additional unique challenges to the timber truck, which needs to be considered. Given these circumstances, the favourable conditions, good contacts to key actors and data accessibility

through researchers at KTH, and one timber truck up and running, it was decided to focus on this specific area named site 4B.

Site 4B consists of several actors built around one forest company. This includes key actors from two timber truck transport companies, and one charging company The transport companies are owned by their haulers which are smaller companies, that in turn are the owners of the trucks and provide drivers. The purpose of the transport company is to facilitate contracts with customers such as forest companies and coordinate logistics between the numerous haulers and drivers. A more detailed illustration of such a network can be seen in Figure 3.1.

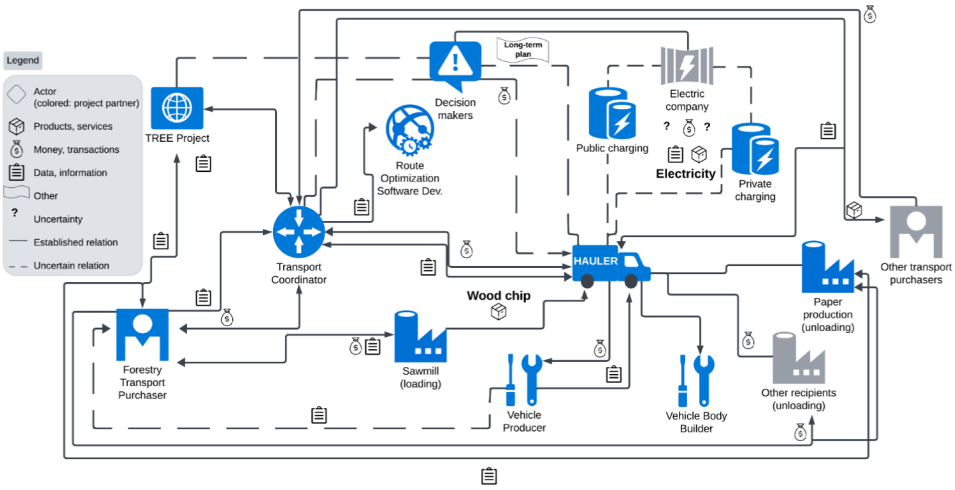


Figure 3.1: Illustration of network of actors and their respective roles (Arekrans, Ölundh Sandström, and Björk, 2024)

3.2 Data Collection

Because of the early stage of the project, it was decided that a qualitative study would be suitable. The purpose was to uncover potential new insights and acquire deep and detailed knowledge to examine this particular case. This was made through semi-structured interviews which enabled gathering of deep knowledge and details (Adams, 2015). Different actors, positions and areas of expert knowledge called for the possibility to adapt the interview guide to further explore the unique research setting. Therefore, throughout the data collection, the interview guide was allowed to be adjusted to fit the informant.

The research started with the study of an introductory conference paper about the TREE project, which was provided by researchers from KTH taking part in this project. Furthermore, one of the authors took part in one of the workshops of the project to get a greater understanding of the different working areas of the project and to get to know some participants. Additionally, the researcher of KTH provided initial contacts to some of the focal actors regarding our scope to set up an introductory interview. The aim of the introductory interview was threefold. First it was used to get more information about the general setup of the project in this specific area, secondly to understand their specific needs and lastly, it gave an opportunity to get more contacts to key actors of this specific ecosystem.

Following the brief literature study and the introductory interview, a first interview guide for the semi-structured interview was made. The information gathered from the literature study, pre-study of the research project and the introductory interview helped form the interview guide in such a structure that would allow the interview to be kept open to new insights, and still touching upon known issues revealed by the pre-study and the introductory interview. The initial interview guide can be found in Appendix A.

The selection of participating interviewees was made by considering the different actors, people taking part in the project and their position in their company. It was important to consider and gather data from all the involved actors of the geographical location and experimental setup to get the perspectives of all actors of the studied ecosystem. A list of potential participants was sent by one of the informants which served as an initial list of contacts. As the interviews were conducted, the informants were asked to suggest additional people to interview. After a few conducted interviews, some patterns started to emerge which revealed additional actors to consider for the ecosystem, which led to a total of nine informants. When these interviews were conducted and analyzed, there was a growing need for external views on both the ecosystem and the resulting insights from the previous interviews. The selection of participants for these later, external interviews was made through counseling researchers at KTH. Three informants were identified from the broader TREE ecosystem with great knowledge about the electric transition and good insight of the focus area. A more detailed representation of each informant and their role in their specific company can be seen in Table 3.1.

Table 3.1: Table of informants for main data collection

Informants inside the ecosystem 4B			
Informants	**Company**	**Position**	**Length**
Informant A	Forest company	Transport development manager	1h
Informant B	Forest company	Transport manager	2x1h
Informant C	Transport company #1	CEO	1h
Informant D	Transport company #1	Transport leader	1h
Informant E	Transport company #2	Sustainability manager	1h
Informant F	Charging station company	Founder and CEO	1h
Informant G	Forest company	Transport leader	1h
Informant H	Research institute	Researcher	1h
Informant I	Transport company #1	Hauler	30 min
External informants			
Informants	**Company**	**Position**	**Length**
Informant J	Truck manufacturer	Part of sales and marketing	1h
Informant K	Optimization software provider	Managing partner	1h
Informant L	Transport company #3	CEO	1h

The semi-structured interviews were scheduled to be one hour long. All were conducted through online meetings. Audio was recorded during the meetings to enable transcribing afterwards. The authors assigned one among themselves to lead the interview, while the other focused on listening and asking follow-up questions or clarification. All of them started with a short introduction from the authors where they consented to being recorded. The informants were given the possibility to look over the results before publishing, to avoid confidential information to be published. Most interviews were kept to approximately 60 minutes. Two interviews were held with one informant, with the possibility to reflect on the data gathered from the first interview and asking questions for clarification of some topics. Table 3.2 provides an overview of the collected data and how they were used.

Table 3.2: Collected data and use in analysis

Data sources and collection (dates)	**Focus and/or content**	**Use in analysis**
Participation in workshop of TREE project (Feb 2025) Participating in an in-person workshop in Södertälje which lasted around half a day. Data consists of detailed notes.	Update on the progress and upcoming events in the project. Work-package presentations and team-based idea generation.	*Complementary data* Provided insights into different research areas within the TREE-project setting covering the whole ecosystem.

Participation in meetings of work package 5 (Jan–May 2025) Mix between in-person and online meetings (MS Teams) that lasted around 30 minutes. Data consists of detailed notes.	Focus in the meetings was on reporting progress in the project.	*Complementary data* Clarified assumptions and updates regarding the project.
Introductory interview with focal actors (March 2025) Informal interview with two individuals with different positions in the same forest company. Data consists of detailed notes. The interview was conducted via MSTeams and lasted about an hour.	Interview set up was to get familiar with the project and the pressing issues. Further contact information to key actors were provided.	*Complementary data* Provided an overview of the project and enabled us to get in contact with key actors within this project network regarding site 4B.
Expert interviews with forest company (March-April, 2025) Semi-structured interviews with two individuals with different positions in the same forest company, whereby one individual got interviewed twice. Data consists of three recordings including their transcripts. The interviews were conducted via MSTeams and lasted around an hour each.	The semi-structured interview guide focused on how the company is affected by the transition and what specific needs they have.	*Main data* Provided insights to the challenges of the transition in the perspective of the forest company.
Expert interviews with transport company (April-May, 2025) Semi-structured interview with four individuals from different transport companies. Data consists of four recordings in English and in Swedish, including their transcripts all translated to english. The interviews were conducted via MSTeams and lasted around an hour each.	The semi-structured interview guide focused on how the company is affected by the transition and what specific needs they have.	*Main data* Provided insights to the challenges of the transition in the perspective of the transport company.

Expert interview with charging provider (May, 2025) Semi-structured interview with one individual from a charging provider which is also part of a transport company. Data consists of one recording in Swedish with its transcript translated to english. The interviews were conducted via MSTeams and lasted around an hour each.	The semi-structured interview guide focused on how the company is affected by the transition and what specific needs they have.	*Main data* Provided insights to the challenges of the transition in the perspective of the charging provider.
Expert interview with hauler (May, 2025) Semi-structured interview with one individual from a transport company. Data consists of one recording in Swedish, including the transcript translated to english. The interview was conducted on the phone and lasted around 30 minutes.	The semi-structured interview guide focused on how the hauler is affected by the transition and what specific needs they have.	*Main data* Provided insights to the challenges of the transition in the perspective of a hauler using the electric truck.
Expert interview with external actors (May, 2025) Semi-structured interview with one individual from a transport company, optimization software provider, and truck manufacturer each. Data consists of one recording each, including their transcripts. The interviews were conducted via MSTeams and lasted around an hour each.	The semi-structured interview guide focused on how the company is affected by the transition and what specific needs they have.	*Main data* Provided insights to the challenges of the transition from an external perspective, which aimed to provide rigor to the analysis.

3.3 Data analysis

Semi-structured interviews have its aim to provide rich qualitative data. Limited amount of interview possibilities but in-depth and possibly novel insights required a method of analysis to further embrace the exploratory approach. The resulting choice of structure for data analysis was inspired by Corley and Gioia (2004) and Eisenhardt (1989). The initial literature study gave an initial insight to the type of result that could be generated by the case,

and what theories could be applicable which somewhat resembles the method of Eisenhardt (1989). Still, to keep it as explorative as possible, the coding of the data was done according to that of Corley and Gioia (2004), starting with open coding. The extracted parts were grouped by specific topic and implication, which led to a number of 31 first order codes. The relationships between the codes were examined and then grouped into 12 second order themes. Lastly, these higher order themes were grouped and categorized into five aggregated dimensions, which would represent the basis for connecting the results to theory. The final aggregated dimensions consist of; Project serves different organizational purposes, Haulers depend on forest company risk taking, Resistance to changes, Decision-making shifting inside transport companies, and Technological barriers are a bottleneck to the transition. Figure 3.2 gives a visual representation of the coding tree with some of the first- and second-order themes, as well as the aggregated dimensions.

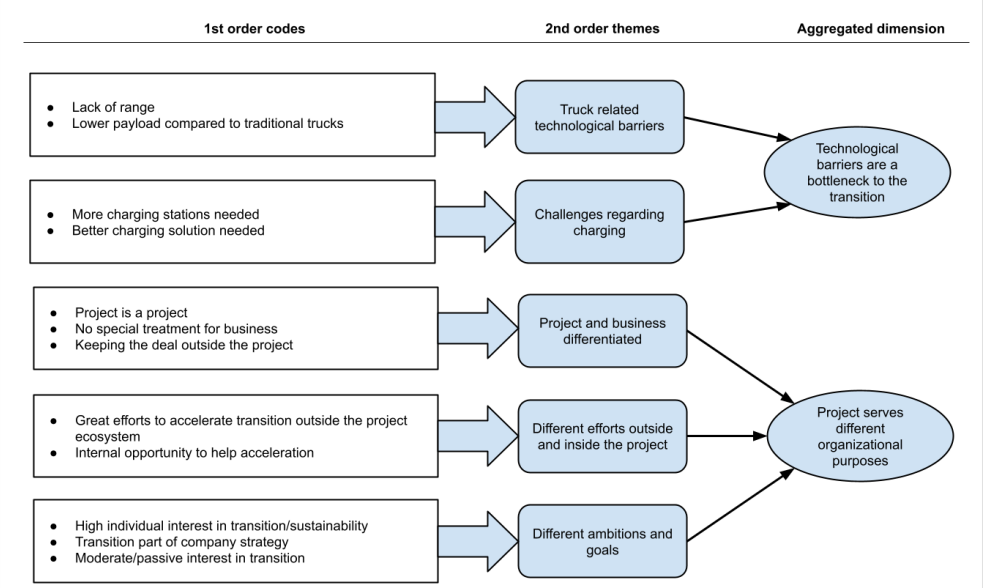


Figure 3.2: Visual outtake of the resulting coding tree

The external interviews were analyzed with the purpose of confirming or rejecting the aggregated dimensions and themes, as well as finding new insights and perspectives. This was inspired by Eisenhardt (1989) as a way of adding rigor to the research. The data analysis of the external interviews was thus conducted separately to keep actors strictly within the ecosystem of site 4B apart from the ones who are only partly involved. The purpose of this was to be able to use the "external" interviews more effectively as complementary

data and serve to analyze the results of the case more accurately. Confirming, rejecting and new insights that were related to the themes of the previous data analysis were extracted and categorized by informants and presented by general topic.

3.4 Methodological evaluation

The credibility of the study was ensured/enhanced by the usage of several data collection methods, including external interviews and complementary data. Despite not being a proper triangulation, it does provide some credibility to confirm the findings. Further, there were a number of informants used who were more or less independent from each other. Several of them are part of the same project and some are part of the same company. However, the number of informants, the length of the interviews, their different formal positions and the fact that they were interviewed at different occasions add some credibility to the results. The research process is well-documented and could be followed for a similar setting. There is a good potential of transferability as the method used is applicable on a conceptual level. The roles of the actors are fairly well described and the context clear. With the choice of data analysis, the findings can easily be traced to a specific actor or actors, giving the possibility to apply the findings to other contexts where they seem fit. The conducted study is limited to a very early stage of the project, but during the conducted study, very little progress was made in the project itself. Although the potential transferability is reduced due to the specificity of the phase of the transition, it is increased for other settings in a similar stage of their sustainable transition.

The point of time that each type of data was gathered is quite clear, although somewhat vague relative to each other by only providing the months they were relevant. The exact dates were not considered relevant enough to include and focus was put on providing clarity on what data was collected, and how it was collected, rather than exactly when, which would add complexity to potential replicability. To the advantage of the methods used here, the total span of the research is quite clear and short. A clear record has been presented of the amount of informants and their types, the extent of the interviews and the findings derived from each interview through the design of the data analysis. This gives the research dependability. Further, the data analysis provides high traceability of the results which provides confirmability.

Chapter 4

Results

This chapter will present the five aggregated dimensions derived from the data analysis.

4.1 Technological barriers are a bottleneck to the transition

The interviews revealed that the BEV truck is clearly lacking in performance compared to the diesel truck. Prominent barriers are lack of range, long charging times, and lack of charging infrastructure were mentioned by all of the informants. These technological barriers pose a significant challenge to the transition. 4.1 provides an overview of the aggregated dimension.

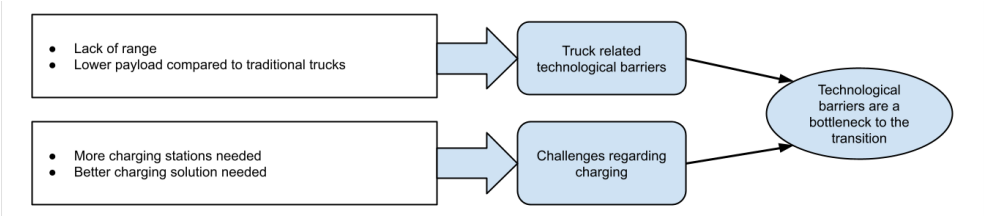


Figure 4.1: Illustration of first aggregated dimension, including second order themes and first order codes

4.1.1 Truck related technological challenges

The results of the interviews revealed a clear consensus on the technological barriers of the BEV truck. Informants from the transport companies, forest

industry, charging provider, and the hauler all agree that the current electric truck is clearly underperforming compared to the diesel truck. The lack of range seems to be the most prominent barrier. While different incentives and ideas to get around this issue were investigated by the actors, it still seems to be an unavoidable bottleneck to the transition. Informants from the transport industry specifically mentioned that the lack of range was the biggest burden to the use of the current route planning methods. The problem of the lower range significantly influences the distances you can go with the electric truck and requires extra attention to observe the battery levels and road conditions. This resulted in the perception of the electric truck being less flexible than the diesel truck by the informants. The BEV truck is therefore bound to certain routes, which can create tensions with other haulers. Despite that it was mentioned by some informants that the truck works very well for shorter distances, for a proper adoption of the truck there is still the need to extend the range of the BEV truck to be comparable to the traditional diesel truck.

> *"It's all about how far you can go with the truck."*
>
> — CEO, Charging company

Many informants also raised concerns about the difference in payload an electric truck can handle. Due to the heavy battery in an electric truck the overall payload needed to be reduced in order to compensate for the overall load allowance of the whole truck. This significantly influences productivity.

> *"And then you have also that you lose some payloads because those [electric] trucks, if we have the same total weight, you lose about 1.6 to 2 tons payload. You shoot it right by the door with the payload. So they are not so productive in that way."*
>
> — Transport development manager, Forest company

4.1.2 Challenges regarding charging

A great focus during the interviews were put on the charging infrastructure. The availability of charging stations was something that some of the informants mentioned as an issue. This was despite the fact that the site in question has one of the best charging networks in the country. The main problem the informants raised was the fact that timber trucks drive in the forest, where there are no charging stations. Another operational barrier is that the harvesting locations are never the same, meaning that the distances and routes

will be constantly different. For this reason, the extensiveness of the charging infrastructure is extra important. The forest company sees a possibility to build charging stations on the land of their industries. This however, can pose some challenges as one informant highlighted.

> *"But also, if you would optimize this, you should charge at the industry where you unload. And a lot of them are not able to have a charger because of the lack of electricity of enough locations."*
>
> — Sustainability manager, Transport company #2

Another efficiency gain for the charging network would be to charge while unloading, which the transport leader from the forest company saw as an opportunity. Except for the density of charging stations, the time it takes to charge is also important and currently insufficient according to many informants. The transport development manager mentioned both the actual time it takes to charge and the waiting times by the charging station as reasons why charging takes too long. At the moment, the waiting time is a non-existent problem according to the charging company CEO, but says that more traffic might create a need for a booking system. The informant also speaks about faster charging technology, but adds that no trucks are equipped with technology to utilize that yet. The transport leader requests that kind of charging technology and concludes that the electric truck will be a mere substitute to the diesel truck if a full charge takes 30 minutes.

> *"From a very empty battery to a full battery, in order to be competitive with a regular truck, you have to let it rest for about half an hour. It should be held together with the breaks that you have to take while driving."*
>
> — Transport leader, Transport company #1

Furthermore, considering a scale-up of the BEV trucks, many informants revealed that the capacity of charging stations might face a significant challenge to accommodate several trucks at once, which already is a challenge in some cases, as described by a driver.

> *"If you're going to share the electricity with each other. Then it takes too long."*
>
> — Driver, Transport company #1

4.2 Project serves different organizational purposes

The aggregated theme "Project serves different organizational purposes" shows that there are differences among the actors in the ecosystem in terms of the commitment they're making in the transition towards sustainable business models. Some actors are spending a lot of resources on building new business models around the new technology. Others are more focused on not missing out on the potential opportunity. The project thus serves different purposes for the different individual firms in the ecosystem. For some actors it serves as an important but almost complementary venture in the journey towards a transition, whereas for others the research project seems to be one of the more advanced undertakings in their organization. This gives indications on how the attitude towards the transition and the project might affect the success of the transition in the ecosystem. Figure 4.2 gives an overview of the aggregated dimension.

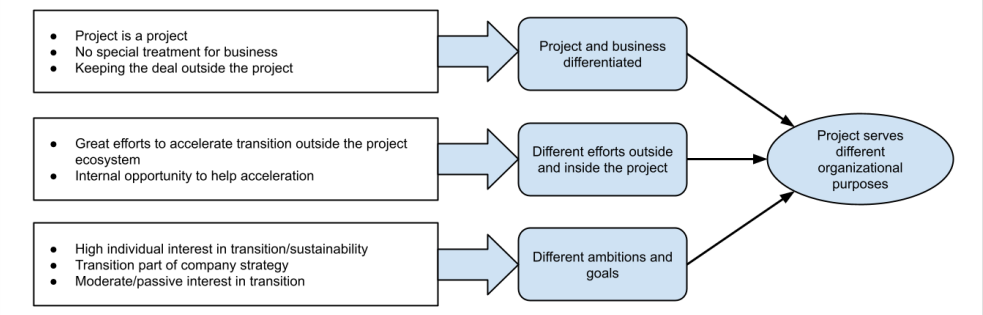


Figure 4.2: Illustration of second aggregated dimension, including second order themes and first order codes

4.2.1 Project and business differentiated

Some participants were keen on highlighting that the research project is just a pilot project and not an actual business. A transport leader from a transport company mentioned that the transport company owning the electric truck was a decision taken purely for the project, and that it might as well be a hauler that owns the truck in the future, emphasizing that the business setup in the project should not be viewed too formally. The transport development manager also emphasized that the project is a pilot where testing is the main priority, and

that they should start slowly. It is clear that most actors think of the research project as just a project and not a means by which the transition will take place. The setup is temporary and likely needs more time to be evaluated.

> *"A project is a project."*
>
> — Transport leader, Transport company #1

The CEO of the transport company shared a similar view on the research project. It is supposed to be seen as research, and not business. Transparency should be high inside the project but limited outside of the project. The transport company was having discussions with the charging company about electricity prices, but had that only for the purpose of the project. The CEO of the charging company spoke about the project as a possibility to test the truck and setup but sees the real proof of viability outside the project, through normal business. The relationships seem temporary as well. While many of the actors do have business with each other outside the project, which the interviews have revealed, the way they conduct "business" in the research project is not to be taken as a viable business for the most part. The two transport companies operating the electric trucks are unsure about their ownership model, where one of them is owned by the transport company, and the other is leased. Both of the companies were very reluctant to speculate on whether any of the business models would enable haulers to acquire electric trucks.

> *"But in the long run, the electrification has to be on its own, so that it is competitive. Otherwise those products will not sell in the future."*
>
> — CEO, Charging company

All the current business regarding electrification is kept outside the boundaries of the project. The CEO of the charging company was very specific in describing the collaboration with the other actors in the project. The forest company did not receive any special treatment in terms of placement of charging stations. The only consideration was the amount of traffic, harvesting sites delivering areas, and then the charging company alone took decisions about when and where a charging station would be built. Most other industries were considered in that decision, indicating that the research project is not seen from their side as a means of acceleration in itself, but rather a learning opportunity. No charging stations were built only for the purpose of the project. Most of the actors seem to agree that the research project serves as a testing ground where the initial ideas should not be taken too seriously yet.

4.2.2 Different efforts outside and inside the project

The differing purposes of the project for the actors is also shown by the efforts they make to expand their network and ecosystem outside the project. The forest company revealed that they are trying business model setups for semi-public charging stations outside this ecosystem. The purpose is to figure out how they each influence the price for charging electricity. On this site only public charging was used. In the other site they are building charging stations on their own land, in proximity to their facility, to facilitate their suppliers. They are asking themselves what the best business model setup is, and if they should own the charging equipment and setup as well, or rely on a third party. The forest company is doing a great deal by investigating different business models regarding semi-public charging.

> *"But in another site, we are trying other business models where we, together with some partners, built up a system. You can call it power as a service, someone else is putting the equipment in our mill area, so to say, and then we are able to set the price for the electricity."*
>
> — Transport development manager, Forest company

The charging company's business already relies on the technology considered belonging to that of the transition. They are building many charging stations and are planning to acquire already existing ones to further expand their network. Their situation is slightly different compared to the other actors in the ecosystem, in the sense that they are focused on this transition. Even though one might argue that the charging company is already ahead of the others in time, they do rely on this industry, among others, to make the transition. The charging company CEO mentioned that the usage of the charging stations is so little that a booking system is not required yet, meaning that the utilization of the charging stations are only a fraction of what they could be at a future stage. Thus they are taking a great risk when building new charging stations, especially since their whole business relies on this transition.

> *"It's not a need right now. There is a lot of capacity compared to the number of cars."*
>
> — CEO, Charging company

Some transport companies are experimenting inside the TREE-project scope, for example, setting up charging stations at their hauler's homes to

lower costs for charging. This was considered crucial for the execution of the project. Such initiatives represent the individual efforts to aid the transition of some actors. Another of the transport companies owns the electric timber truck used for the experiment, which is not according to their normal business, but is their greatest contribution. The residual value was mentioned by many as a great uncertainty, but the value of ownership was deemed greater.

4.2.3 Different ambitions and goals

There were some differences in goals and interpreted ambitions between the actors of the ecosystem. In relation to their efforts and view on the project, the same actors who were seen as the least ambitious also had the least efforts outside the project.

One of the transport companies has set clear goals for sustainability in different steps. They align with the goals of the project and other federal goals for sustainability, which is to buy 50% non-fossil trucks by 2030. Almost all respondents considered the site to have one of the best conditions in the country when it comes to charging infrastructure. The CEO of the charging company was also part of starting the research project. The forest company had the long-term sustainability goal of being fossil free by 2035. The transport manager concluded that the project and transition is crucial for the company to reach their goals, which is to lower their CO2 emissions by 50% by 2030. It also helps the forest company reach their purpose of doing good for the planet. All these actors show high ambition and goals to make the transition.

> *"We have a part goal that we will be 50% not fossil fuel. 50% 2030 and 100% 2040."*
>
> — Sustainability manager, Transport company #2

The other transport company saw the transition towards more sustainable business models as a possibility to increase their revenue. They did not have a set sustainability goal but expressed that it could be a new business opportunity. The feeling was that this transport company was slightly sceptical of the electric transition in general. This was justified by the company being much smaller than its competitors and wanted to wait for the results of the project before taking action.

> *"We haven't set up that yet but it's very clear for us that the customer demands work in this area and you have to show them that you are into it."*
>
> — CEO, Transport company #1

4.3 Haulers depend on forest company risk taking

The interview data revealed that transport companies are willing to support the transition, but are mainly hindered by their dependency on their customers, the forest companies, to take the leading role in both financial and operational responsibility. Figure 4.3 gives an overview of the aggregated dimension.

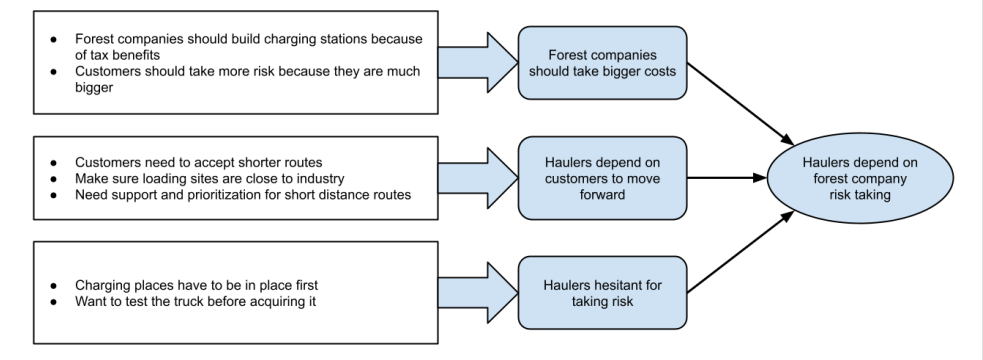


Figure 4.3: Illustration of third aggregated dimension, including second order themes and first order codes

4.3.1 Forest company should take bigger risk

A common theme among the interviews with the informants from transport companies was that the forest companies are naturally better suited to carry financial risks of the electrification due to their significantly bigger size, available capital and ability to access subsidies. The CEO of transport company #1 underlines the significant difference in financial resources by mentioning that spending time on the project is already a risk for the transport company, emphasizing on the tight margins of transport companies.

> *"The risk we take is all, I'm sitting here, not for free, we're not charging my cost, we're not charging our organization's cost, and so on, so we invest in the project also, but we compared to [Forest Company], we are very, very small, they are very, very big, so that's quite natural that they take the risk."*
>
> — CEO, Transport company #1

Informant E expressed a similar view on this issue. High initial costs were perceived as a prominent challenge, especially for the transport companies' haulers, which reflects the need for sharing risks accordingly. Moreover, informant E mentioned that a higher interest by their customers, the forest companies, would help them make the commitment. Furthermore, informant E expresses the need to collaborate more with their customers to prepare them and that they are willing to join. This was supported by one informant addressing the circumstance that the forest company is driving the transition and are therefore expected to pay more for the fossil fuel free alternative.

> *"At first, they would have to pay more than fossil fuel."*
>
> — CEO, Charging company

Insights from interviews show that there is a great need for expanding the charging infrastructure. It was mentioned that some efforts have been made from the forestry company to set up semi-public charging stations at their site. These kinds of charging stations are accessible to all the transport companies supplying the industry, which also means they cover costs and therefore mitigate the risks for the transport companies. However, one interview with an informant from a transport company revealed that this kind of support was taken for granted, indicating the expected risk distribution across the actors based on their resources. This view was not shared by other informants from the transport companies. Informant C saw it as an opportunity to possibly lower the taxes of the energy prices when charging stations are set up by the industry themselves, raising the need for the forest company to act upon that in order to bring down the electricity costs.

> *"Is it possible to recharge the truck within the industry and then get rid of the taxes?"*
>
> — CEO, Transport company #1

4.3.2 Haulers depend on customers to move forward

Transport companies clearly depend on driving steps from forest companies to make electric transport viable. One of the major differences of a BEV truck to a diesel truck is the significantly lower range. This comes with a lot of limitations to the usual operations of a traditional diesel truck. Within the project, the actors try collectively to target this challenge, however, the interviews show that especially the transport leaders from transport companies

and their haulers are mostly affected by it. Despite the promising observations that the truck operates in shorter distances almost like a diesel truck, as most of the interviewees mentioned, it depends a lot on the forest companies to provide these shorter routes and therefore, their support.

Many informants noted that successful implementation requires careful planning to ensure routes remain short and that charging is within feasible distance. Informant C emphasized that current route planning heavily depends on customers' cooperation.

> *"We need all their support, and we get it, we are prioritized regarding the harvesting sites."*
>
> — CEO, Transport company #1

Furthermore, the CEO emphasized that the truck needs "to keep rolling as much as possible, and therefore we have loading sites that are quite close to the industry." However, this strategy also comes with trade-offs. While BEV trucks can have secured short distance operations, traditional diesel trucks need to cover more long distance routes. This can consequently raise conflicts between electric truck operators and diesel truck operators, since short distance routes can be useful to increase productivity of truck transportation regarding filling loads, especially when the driver's time is limited. Informant E highlighted that they need to work with their customers to solve this problem.

> *"It's necessary for us because it's so much money… we need to work a lot with the customers."*
>
> — Sustainability manager, Transport company #2

Informant G confirmed this issue, by revealing that more communication was needed with other hauling companies to find compensation models to reduce the tensions between other haulers for taking substituted routes. In the current state of the project only one transport company was operating with a BEV truck, however not in their usual transporting area. This, however, makes operating the truck even more challenging to compensate for, because their own company can not compensate for it with additional trucks.

Another burden is the limited charging availability. Despite the efforts of the charging company to set up more charging stations, it is crucial to have those within reasonable reach to the transport destinations. Informants from the forest companies identified the need to set up charging stations at their

locations to allow more possible routes for the transport companies. This however, indicates again the dependency of the haulers on the forest companies to move forward to get more available routes.

> *"If there was a charging station in all these larger industries, it would be completely different."*
>
> — Transport leader, Transport company #1

Another aspect is that the current strategy for route optimization relies mostly on a software, as some informants of transport companies revealed. The software showcases all the wood flows from different customers which need to be transported. This includes the stock at the harvesting area and how much from it needs to be transported to a certain destination. Haulers can see themselves and others in it and use it to share loads and optimize their routes to utilize the truck as much as possible. However, some informants mentioned that for better route optimization it would be necessary to have all the customers in this software to be able to also optimize for shorter routes for the electric trucks.

4.3.3 Haulers hesitant for taking risk

A common theme across the interviews was the hesitating behavior of the participants from the transport companies. The root for the hesitating behavior can be traced back to the high initial costs and uncertainty of this transition. The issue with the interdependency to compensate for the lack of range with more charging opportunities, which is currently also lacking to some extent, raises the question by some informants on what has to be there first for a successful scale up. This "hen and egg" problem is reflected throughout some of the interviews. While one informant of a transport company expresses the need to have the charging places in place first, the CEO of the charging company mentioned that new charging stations will only be built if there is enough business to do. The hesitant behavior of the transport companies and haulers was identified by the forest company. Consequently, the transport leader of the forest company mentioned that it was necessary to show that it is economically possible to do the transition.

> *"So, if we can show that it's economically beneficial, then I don't think it's a problem."*
>
> — Transport leader, Forest company

One of the transport companies mentioned their haulers, by which they are owned, and their attitude towards the transition. One of the informants from the transport companies expressed a reluctance among the prominent persons from the hauling companies/haulers about acquiring an electric truck. Some of them wanted to test the truck themselves before deciding if their drivers should even be educated in driving electric trucks, indicating that the technology and economy has to be in place before they invest. Other haulers have shown great interest in the transition, according to the transport companies and the forest company.

4.4 Resistance to change

The transition to BEV trucks requires rethinking of how the truck should be operated to make the transition viable, due to the limitations of the BEV truck. Range, less developed infrastructure and high initial investment represent the main burdens which need to be overcome. However, the interviews reveal that not all actors are ready for such a change. The resistance to change hinders insights to new possibilities to make this transition viable, and the possibility to miss out on some potential opportunities to change the business model in a completely new way to make it more viable. Figure 4.4 gives an overview of the aggregated dimension.

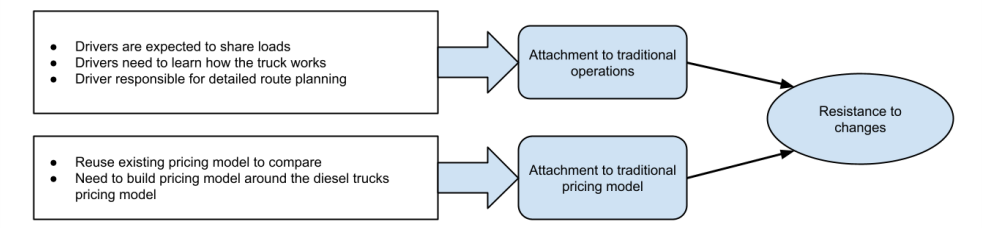


Figure 4.4: Illustration of fourth aggregated dimension, including second order themes and first order codes

4.4.1 Attachment to traditional operations

Throughout the interview, informants persistently highlighted the significant difference in performance of the BEV truck compared to the diesel truck and how those affect their current operations. Many expect electric trucks to reflect the performance of diesel trucks, which, though, is hindered by the

underperforming technology of BEV trucks. However, interviews indicate that some transport companies are still locked into their traditional planning methods, trying to fit in the electric truck. This indicates that a lot of efforts within the project have been made just to make the electric truck align more with their traditional operations. This however, comes with the cost of the need to move around additional weight, which results in an even worse payload, which leads to even less productivity. This clearly indicates that a rethinking of suitable operations for the BEV truck is needed.

Traditional operations heavily rely on the drivers ability to plan their routes by themselves. With the new electric trucks they are still expected to do so to some extent. The experience and ability to be dynamic in their planning is very important in today's business model. This however, is now challenged by the significant difference in performance of the truck. Interviews with several of the actors revealed that the drivers of the truck do not seldom exchange loads with each other. The sustainability manager described this as a way for the driver to optimize operations. The forest company expressed understanding and said they expected suppliers, and drivers specifically, to exchange loads intended for other forest companies. This shows a great trust and responsibility placed on the driver in the operation. The haulers are working with extremely low margins, indicating that the exchange of loads might be one of the deciding factors for profitability. The driver creates a network so they can manage this effectively:

> *"And I would say that it also enables an optimization that is not possible when we only take our own wood in consideration, because the drivers can create a network where they exchange loads with each other, including other companies as well."*
>
> — Transport manager, Forest Company

The importance of the planning being on driver level was mentioned by multiple participants, which makes it seem like the experience and competence of the driver is key in the planning of the routes currently. One of the transport managers explained that when conditions are favourable, communication can be as infrequent as once per week. It was only in difficult situations when communication between the transport leader and driver was needed. But several respondents mentioned that even in those conditions, when the weather and the forest roads are bad, the ability of the driver to respond to the changing conditions and change the plan and route is key. The autonomy of the driver thus, seems really prominent, and also seems to be valued by themselves:

> *"...so that is why it's almost necessary for some of the planning at least to be on driver level, and that is also a way for them to also control their own business. To set up routes that are good for them economically."*
>
> — Transport manager, Forest Company

4.4.2 Attachment to traditional pricing model

Another burden closely connected to the operations is the pricing model for the trucks, which is set by the forest company. Again the currently underperforming technology of BEV trucks requires new ways of thinking for the pricing models. The current pricing model is set by factors like downtime, transport distance, which covers fuel consumption, wear, and service, and the amount of tons which needs to be transported. While the BEV truck can outperform the diesel truck in some factors (e.g. requires less service), it is still a major challenge to take the negative aspects of the new technology in consideration. Differences in payload, lack of range, and long charging times, all bring challenges to the current pricing model. Especially, the fluctuating electricity prices play a crucial role, as emphasized by the transport manager of the forest company. The current pricing model accounts for fluctuating fuel prices through a special fuel compensation model. It consists of a baseline of the fuel price and if that varies from month to month, the price of the logistics will be adjusted accordingly. To test the trucks performance, one informant revealed that they still use the traditional pricing model in order to compare it better. However, the fluctuation of the electricity prices is significantly more dynamic, posing a challenge to this model. However, actions to undergo a holistic change of the pricing model seems missing. The currently operating truck is using the same pricing model as the diesel trucks to compare the costs, according to the CEO of the transport company. Similarly, the transport manager of the forest company indicates the thought of replacing the pricing model for the BEV truck with the traditional, but automated model.

> *"Well, something that I have been asking for is help to find an automized compensation model for electric trucks, the same way that we and the rest of the business is working with diesel trucks."*
>
> — Transport manager, Forest Company

4.5 Decision-making shifting inside transport companies

Planning is currently quite decentralized in the timber hauling industry. Informants highlight the value of their competence and experience when making decisions in the day-to-day operations. As electric trucks are introduced and the transition scaled up, this will likely change, according to many informants. The planning will most likely be managed more digitally and fall to the transport leader. On top of that, the charging infrastructure is lacking according to many, which requires new ways of working. Figure 4.5 provides an overview of the aggregated dimension.

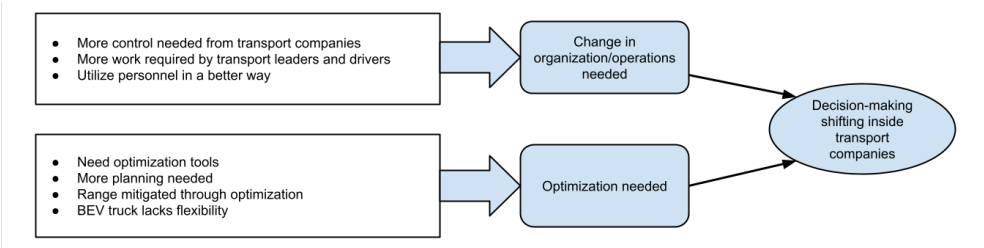


Figure 4.5: Illustration of fifth aggregated dimension, including second order themes and first order codes

4.5.1 Change of organizational operations needed

A recurring theme among most informants has been that transport companies need to take more control over operations. Currently, most planning is done on the driver level. Several informants indicated that the operations at the transport companies need to be altered, and that more control is needed from the transport companies, meaning the transport leaders. This would indicate a change in the whole business model of the transport companies, in the sense that the way they work is much different from before. The CEO of the charging company mentioned as an example that they need to conduct their planning considering things such as avoiding energy spikes. The informant highlights that there will be a greater need for information, and that the electric trucks need to be controlled in a different way. The sustainability manager held a similar reasoning and concluded that it is the transport leaders and drivers that are mostly affected by the whole transition. The charging company CEO also

summarized the main challenges of the transition, where control was one of the parameters:

> *"It's about finding business models, charging infrastructure and control."*
>
> — CEO, Charging company

With the changing operations and structural changes, some informants naturally highlighted that it will require more work from transport leaders and drivers to conduct the planning. If there is a greater need for information, that would require more time to process that information. Since more planning is placed on the transport leader than before, that would also imply more work in itself. The transport leader explained that in their company, there are currently three transport leaders handling around 100 trucks in the transport company. The sustainability manager talked about optimization and the need to follow the soon-to-be introduced truck live, which would require constant planning:

> *"And we have transport leaders here who will check, follow that car and plan for it all the time… it will demand time from the transport leader."*
>
> — Sustainability manager, Transport company #2

4.5.2 Optimization needed

Currently, the electric timber truck performs significantly worse on longer distances than the diesel truck, which is one of the main bottlenecks. While diesel trucks have the opportunity to take longer transportation routes, electric trucks are bound to the ones close to the industry, which indicates that this lack of flexibility is disrupting their traditional route planning strategy. Informants indicate that by mentioning that range and the insufficient charging availability limits the electric trucks only to dedicated routes, which hinders their traditional way of working. Example provided from an interview with a transport company, highly dependent on their drivers for planning the detailed routes, expressing the fact that:

> *"[The drivers'] work is not as flexible as a traditional truck …it requires more planning."*
>
> — Transport leader, Transport company #1

To get around this, different kinds of optimization have been suggested and mentioned by informants. Tools for optimizing by taking all factors into consideration for example, have been mentioned as a solution. The transport development manager also talked about optimization tools to enable booking slots at charging stations, and plan the amount of energy needed. The transport leader at the forest company also mentioned the queues at charging stations as a possible problem, and optimization as a solution to that. The CEO of the transport company revealed that they will look at optimization later in the project, and compared the need of an optimization tool to the car industry, where Tesla were the first to offer a complete solution:

> *"Then I changed to Tesla. There you have the solution, it's a network. You don't buy a car, you buy a solution… you push a button and the system says, all right, go there, there and there, stop there. It applies to when you are allowed to drive and when you have to take your rest… When we are there, then we have a good solution. Then I think electrifying even our difficult trucks will be successful."*
>
> — CEO, Transport company #1

Moreover, the amount of planning done will likely be greater. In the beginning, the transport leader of the forest company had daily contact with the driver of the electric truck. They chose routes that were suitable both in terms of distance and charging availability. The transport leader emphasized that the charging possibility and range constantly need to be taken into consideration by the driver and transport leader. The informant also expressed scepticism towards optimization tools, for the simple reason that it does require more work. On the question of which is the biggest challenge at the moment, the transport manager answered that the additional planning is the greatest issue, and especially when scaling up. The transport manager described it as follows, referring to geographical areas as challenging for planning:

> *"But I would say perhaps the additional planning then, that it takes a lot of additional work to plan around this and find these harvesting sites that suit the truck. And of course, if we scale this up, I think that we have some challenging areas."*
>
> — Transport manager, Forest company

4.6 Insights from external interviews

Below follows the data analysis from the external actors interviewed at a slightly later stage in the study, presented by informant.

4.6.1 Optimization software informant

Optimization is a crucial thing for the transport companies, according to the managing partner. The timber industry is very decentralized, which needs changing if they want to transition to BEVs. The users of the optimization software will be transport leaders at the transport companies. Their manual planning will be partly substituted with digital and automatic planning, but the managing partner emphasizes that they will still be needed. This will make the driver's day-to-day operations much more strict. These are all steps toward more centralized planning, which will make transport companies more competitive according to the managing partner.

The managing partner mentioned a few barriers to centralized planning and optimization. To be really effective, optimization should be done on a big scale. The managing partner mentioned it as the "third dimension", but also noted that it is not possible at the moment. The different transport companies, who are competitors, own the transport layer which hinders optimization on a bigger scale. All haulers of the transport buyer need to be in the same system in order to effectively optimize, which they are not at the moment. In the beginning, small transport companies would not need to optimize either, since they are so small.

Digitalization is needed generally in the forest industry/timber industry/logistics industry, according to the managing partner. To succeed with optimization, there needs to be data available, of which there is not enough today. With good data, the optimization can give much better results, also compared to manual planning. The managing partner highlights the transport companies especially as the ones who need to digitalize. There is also an opportunity for extra income for the small transport companies if they own some charging infrastructure, and can share that with other haulers.

Small transport companies are taking a big risk though when investing in electric vehicles. The managing partner thinks a sharing economy could help drive down the cost of electricity and help the transition. The industry needs to own the infrastructure themselves in a long-term perspective, according to the managing partner. Forest companies can change their business models to offer charging at their sites and only pay for the transport itself. They should drive

and invest in the transition and can benefit greatly from optimization, says the managing partner. Range is not really a barrier anymore, and the technology development will continue to advance. What is missing now, according to the managing partner, is government support through higher diesel prices, to make the electrification more attractive/beneficial.

4.6.2 Transport company CEO

The CEO of the transport company really highlighted the diesel price as a big barrier for the scaling of electrification. When the project was started, the prices were at much higher levels for diesel, making the electric more attractive. Bringing charging costs down is not enough, according to the CEO. The diesel price must go up if the electrification is to accelerate. But when it does, the transition will be quick. At the moment an electric truck is 50

Route optimization and digitalization was mentioned as a barrier for scaling the electrification. The CEO mentioned that manual planning will be overwhelming for transport companies when introducing more electric trucks. To make the transition and have a substantial part of the fleet as electric, optimization is needed. Support systems for the transport planners have to be introduced to handle the amount of electric trucks. Central planning can also bring great efficiency gains to the timber hauling companies, who are currently very much relying on the drivers to do the planning. With central planning and optimization the whole operations can get much more efficient, which goes together with electrification, according to the CEO. Haulers switch loads with each other, but can be selective in their choosing, which is proof for the CEO that the competition is not good enough in the timber hauling industry, or the pricing is too high.

The CEO mentions the charging infrastructure and deems it sufficient for electrification at the moment and for their operations. However, the informant points out that it might become a problem in the future. It is important for the timber hauling sector to map out charging stations to ensure that the future demand is met. The technology is good enough, according to the CEO, it is just the economic part that needs to be figured out. Apart from the diesel prices, the residual value was also mentioned as a barrier economically. Haulers are taking the biggest risk in this transition when they invest in electric trucks. Even if the residual value is high and the performance of the electric truck is good, the total costs are still quite high. The transport company is working with the truck manufacturer to find an application for the second life of batteries.

The transport company works a lot with their customers who are now quite

positive overall towards electrification of their fleets, but the CEO points out that it depends on where they stand in their process. Despite the difference in costs between BEVs and diesel trucks, customers are positive towards BEVs. The same thing goes for their haulers, who are now asking for electric trucks, which has not always been the case. A lot of work has gone into convincing haulers. They are very important for the transition, and big haulers have shown to be extra important for the transition so far.

4.6.3 Truck manufacturer sales worker

The sales worker for the truck manufacturer saw the charging infrastructure as the main barrier for scaling the electrification, but also mentioned some solutions to that problem. The grid power was one of the problems, which the forest companies could get around by setting up charging at their sites, where the grid is already very strong. That would also give them the opportunity to control the costs of charging for their suppliers. Moreover, more collaboration in general is needed regarding charging infrastructure to make sure it can meet the demand, according to the sales worker. However, it will be difficult since the owners of the charging infrastructure are competitors. If the charging infrastructure is not sufficient, other "fuels" will have to cover those distances, which might become a problem if there are many electric trucks in a fleet. The sales worker says that there will be a mixed fleet in the future, but that BEVs will be a majority. Most transport distances should be reachable with the BEV, even if the range is currently a concern. The sales worker is optimistic about the rate of technological development for the coming years.

Sustainability is an added value and can give the forest company a competitive edge, according to the sales worker. The value has to be clear though, in order for it not be viewed purely as a cost. It is possible that the end-customer has to take the cost for sustainability, the sales worker mentions. Not many companies are willing to take an extra cost though, even for sustainability. There is still a big focus on the costs though for the forest company, but transports should be a low-hanging fruit to address regarding sustainability. However, there is a risk for the hauler, since the BEV is around three times more expensive to buy than a diesel truck. Especially if there are small haulers. The length of contracts between transport buyer and transport company can really help in reducing that risk and accelerating the transition, by making them longer.

The financial sustainability has not been discussed too much in the project, but needs to be figured out, the sales worker says. All the actors need to work

together to make the transition successfully. The forest company is the actor who should be driving the transition though, and put demands on the transport companies. The truck manufacturer and the forest company are also starting to collaborate/communicate, the sales worker mentions, which is positive. They help the forest company drive the transition through an outside-in position, with the goal to prove to haulers that electric trucks are profitable. The truck manufacturers themselves are quite active themselves in the transition by developing their own optimization software, charging network and charging service.

Chapter 5

Analysis

Below follows an analysis of the results structured to answer the respective research questions.

5.1 Challenges developing business models

Current operations seem well-established and hard to migrate from. The attitude towards the new technology is sometimes focused on replacing the old technology with the new in such a manner that the performance is the same, and business continues as usual with the new technology. This suggests that the solution-space of the old technology is engraved deep enough to result in a cognitive bias towards the new solution (Geels et al., 2017). However, when testing new business models, it can also be a strategy to keep it close to the normal business (Teece, 2018), which seems to be the case in this instance. That would ease the transition from the old to the new, which was one of the identified challenges.

The introduction of the new technology and the potential scaling of it seem to induce some tensions between the old and the new way of conducting business (Van Bommel, 2018). It has been established in the interviews that the new truck operates well on short distances. If that operation should be scaled up, several informants mentioned that as a barrier to the transition, since the amount of short distances are limited and work as “fill-out” to the longer transport distances, to optimize operations. That creates a tension in the operations of each transport company, where the diesel trucks have to be used for the longer distances and miss out on the filling load, and therefore the possibility to maximize efficiency. Another identified tension is found in the designing the compensation model, the cost structure (Osterwalder, Pigneur,

and Clark, 2010), where the new and old technology (and other alternative fuels) should be compensated fairly and equally. It was identified also that this compensation model would need to be automatic, and new competence and resources were called for to develop that.

The experience and competence of the driver is currently seen as a competitive advantage/key resource. With the new technology, the importance of that competence seems to be weakened. Informants were stating that the transport companies were in greater need of information and control and needed support systems due to the underperforming technology and the difficulties with using it in traditional operations, especially when handling multiple BEVs. Systems taking woodstock, route and charging needs into consideration at the same time were suggested as solutions and already partly involved in the project. These support systems would be handled by the transport leaders, suggesting that the decision-making might shift in their operational setup from the driver to the transport leaders, and even further to the optimization software, which the managing partner of the optimization software confirmed. The value proposition of good and efficient logistics is then dependent on the ability of the transport leaders and their potential support systems, and their ability to utilize these. A setup that previously has been characterized as extremely flexible and independent, would now be extensively controlled and monitored. This greatly changes the operations of the transport companies, both in terms of their key resources, activities and competitive advantage (Osterwalder, Pigneur, and Clark, 2010; Teece, 2018) and indicates changes of the user practices of the system (Geels, 2002).

The results of the interviews revealed that the need for new competencies evolved within an organization for the transition. New technology mostly leads to new challenges in new and unfamiliar fields. Barriers related to the lack of charging stations required the project to include charging companies in their ecosystem. However, due to the persisting high costs of public charging stations and the lack of charging stations in favourable strategic locations, the forest company and the transport companies are now forced to react with their own competencies on it. Thereby, informants stated that charging stations at their facilities could be the solution to this issue. However, necessary capabilities are clearly lacking within their respective organisation to make this happen. This requires them to form new partnerships in order to get access to key resources, which is necessary for the value creation (Osterwalder, Pigneur, and Clark, 2010).

In the pursuit to transition an industry, there might be resistance (Farla et al., 2012). The haulers in this case are showing some indications toward

that. The driving actor, which has been identified as the forest company, should create supportive structures to facilitate change, according to Farla et al. (2012). Building charging infrastructure was identified by both the transport company and the forest industry as such support structures. Optimization was also mentioned by all informants, but not in what way it could be supported. No other support structures were mentioned, and according to one external informant, cutting the price of electricity is not enough to make electrification competitive.

The transition to electrified heavy road transportation poses significant challenges to organizations. New technologies often require new business models which seem some companies within the project are reluctant to. By only holding on to existing ones, added value from the new technology can be overseen. Challenges also occur in transport companies' operations. Due to the lack of range of the new technology tensions arise between BEV trucks and diesel trucks. Furthermore, the dynamic capabilities of some organizations are getting challenged, where some companies need to form a new competitive advantage by shifting decision-making inside the company. Additionally, barriers related to the lack of charging stations might require some companies to form new partnerships and develop new key activities. Lastly, organizations fail to develop supportive structure.

5.2 Challenges with collective value capturing in the ecosystem

The project is showing great collaboration efforts between the actors. Several business models and setups are being tested and information about the experiment is shared between all the actors. This largely resembles that of an innovation ecosystem, which focuses around one technology and the development of that (Adner, 2016). The environment is collaborative and there is a perceived collective value created in the project.

It was pointed out by many informants that this ecosystem was created by the forming of this project and is clearly differentiated from the normal business operations. In some way, that could be seen as these actors trying their respective new business models separately from the normal business operations, which is described by Osterwalder, Pigneur, and Clark (2010) as a common way of trying new business models. As pointed out by Osterwalder, Pigneur, and Clark (2010) and Teece (2018), this needs to be transferred into

the normal business, for which there seems to be no plan at the moment. Since the current setup is dependent on extensive collaboration and transparency and the business outside the project is not characterized by those traits, the results indicate that there might be a challenge in transferring the new business models and relationships into the normal business environment. External interviews emphasized that the collaboration between the actors can become better but that it needs to be there to make the transition.

Testing out in a project environment can also be seen as these actors trying to find a niche market that works before scaling it (Geels et al., 2017). An environment needs to be found where the venture proves suitable, and then scale it. Osterwalder, Pigneur, and Clark (2010) also talks about the need to test business models and prove them worthy before scaling. The current challenge seems to be finding this niche market where the electric timber truck is profitable, to convince sceptical haulers and other customers of its viability. Such a niche market seems to be evolving for short route transportation, but is facing resistance from traditional operations.

The added value of the new technology seems limited. It is stressed by most scholars in this field that the added value and the appropriating of that is the key to a technological shift (Atkova et al., 2025; Chesbrough and Rosenbloom, 2002; Osterwalder, Pigneur, and Clark, 2010; Teece, 2018). In transition theory there is a state called hybridisation which suggests a state in which the old and the new technology complement each other and exist in harmony (Geels, 2002). Finding such a state did not seem to be prioritized, but rather that the new technology would replace and resemble the old one as closely as possible. The plan was to test the electric in a complete fleet of trucks to see how it can perform in a bigger system. If a state of hybridisation is found, multiple business models must be handled (Osterwalder, Pigneur, and Clark, 2010) which was pointed out as one of the challenges scaling up.

In the project, the goal is to reach the commonly set goal and have a competitive advantage when the transition happens on a bigger scale through gaining know-how and finding cost-effective solutions. The collective value of the ecosystem seems to be in staying ahead of the competition in the market. As there seems to be no additional value with the new technology, the focus lies on cutting the costs, which can only be made to a certain extent according to the informants. The ways to cut costs have some implications for the collective value and the formation of the ecosystem. An identified way of keeping the costs down is to keep the value inside the ecosystem. The forest company is testing business models for building their own charging stations, to keep down the cost for charging, which was mentioned by many informants as a

way of accelerating the transition. The charging company does not believe semi-public charging is the optimal solution, but sees the sharing economy with public charging as the way to competitive charging prices. The value is kept within the ecosystem, but to what extent the actors are capturing that value seem unclear.

An identified tension in the system is the competition between the transport companies. While transparency and collaboration is high inside the project, it is quite limited outside. The competition between the transport companies is mentioned by several informants as a barrier. The structure of the ecosystem seems to be strictly hierarchical and competitive between the transport companies (Jacobides, Cennamo, and Gawer, 2018). Their concern is likely the distribution of value in the ecosystem (Adner, 2016).

The forest company needs to take more risk and drive this transition even further for the haulers to join in (Köhler et al., 2019). How they should do that is slightly disputed. Most actors agree that the forest company should give incentives to the transport companies/haulers to buy electricity by being ready to pay more for the new technology and build infrastructure on their industrial facilities to support the electrification. Longer contracts have also been agreed upon, to lower the risk of investment. However, if the collective value and value proposition of the forest company captures limited added value from the new technology, the margins of the forest company either have to be lower or a way of appropriating the added value needs to be found. One external interview highlighted that forest companies are reluctant to pay more for sustainable transports than they do today, but the forest company was also unsure what their customers would be ready to pay for a product produced with sustainable transports. In that sense, even more innovation seems to be needed for cost-reduction with the suppliers, or the forest company needs to work with their customers and value proposition, as suggested by Atkova et al. (2025).

The challenges to creating collective value in the ecosystem are many. First, the value proposition of the focal actor seems somewhat unclear, and places the ecosystem in a state of cost-reduction where competition inside the ecosystem is taking place. The collaboration inside the ecosystem of the project is good but seems to face challenges turning it into business. A viable setup that seems profitable has not been found yet, even if there are promising findings, and resistance from some actors and old business models seems to stand in the way. The old and the new technology are still seen as pure competitors. The project is still in early stages, but needs to resolve these

problems if it is to accelerate the transition.

Chapter 6

Discussion

The aim of this study was to explore the challenges of business model development for the individual actors and their impact on the ecosystem, in a sustainable transition of an industry and the specific setting of electric heavy transports in the Swedish forest industry. The findings showed a number of synergies with the presented literature such as tensions within organizations, the importance of alignment and the persistence of traditional operations which indicates a clear inertia of the industry's organizations for developing business models to accommodate the new technology. Digital tools can aid the transition but are constrained by the deeply engraved user-practices and cultures of the industry. The findings and analysis also highlighted important aspects not fully covered by the literature, which will be discussed below. Additionally, emerging tensions and paradoxes identified during the transition are presented, followed by a reflection on the methods used and their limitations, and concluding with suggestions for future research.

6.1 Importance of digitalization

There were several findings that pointed towards a need for digitalization. It was established that especially the transport companies need to digitalize with optimization softwares in their operations to transition to electric trucks, which would have fundamental implications of their way of working. Digital innovations are often taken advantage of in innovation ecosystems, and can even require that complementary innovations are brought into the ecosystem (Mikl et al., 2020). Additionally, an automatic compensation model was called for by the forest company. Some even expressed that these two innovations were inevitable for making the transition, which raises the question if

digitalization generally needs to be implemented in the ecosystem, or the industry even. Business model development would revolve around other, very different perspectives if that was the focus in the project, perhaps aspects such as sharing data and data quality.

To fully utilize optimization, it should be conducted on a big scale, according to one informant, which would bring added benefits for the forest company. If that cannot be done because of the many small transport companies, there seems to be two ways of approaching the transition. Either to take advantage of the numerous suppliers, or take control of the supply chain. A strategic advantage with numerous suppliers should be the agility of the setup, and opportunity to drive innovation through the competition between the transport companies. The support structures suggested by Farla et al. (2012) might then be articulated through the business model of the transport buyer to further incentivise sustainability practices.

6.2 Role of transport buyer as transition driver

It's reasonable to think that the transport provider wants the transport buyer to bear the additional costs of sustainability, and that the transport buyer is reluctant to do so. A previous study by Dahlgren and Ammenberg (2022) highlights that transport buyers in Sweden specifically have difficulties in convincing the transport providers to switch to a greener technology. They found that transport providers often react to the transport buyers initiative rather than driving the change themselves, which can hinder the transition, which this study indicates as well.

In contrast to our study, Dahlgren and Ammenberg (2022) found that smaller carriers made greater improvements in environmental efforts, which was quite the opposite in our case. It might be the nature of the timber hauling industry which makes the difference, as the flexibility increases with a bigger fleet. Additionally, the investment barrier is likely easier to overcome for bigger transport providers. However, the general perception that smaller organizations are more dynamic supports the study of Dahlgren and Ammenberg (2022). With this project including both bigger and smaller actors, an evaluation at a later stage might reveal how the specifics of timber hauling in Sweden play out in this transition.

Dahlgren and Ammenberg (2022) also stated that longer contracts can have an impact on the transition by giving greater security to transport

providers. All actors in this ecosystem agreed that longer contracts would help transport providers. No consideration was taken into what consequences and implications that has for strategy or business model development. As mentioned earlier, one advantage with having numerous transport companies as suppliers should be the competition between them. Longer contracts might have a negative effect on that competition.

6.3 Emerging tensions and paradoxes

In addition to the aggregated dimensions identified in the empirical results, the analysis also revealed several tensions that actors face during the early stages of the transition. These tensions illustrate how actors are not only facing technical and organizational challenges but also need to navigate strategic trade-offs that shape both business model development and collective value creation. Additionally, two paradoxes emerged which need to be taken into consideration.

6.3.1 Tension 1: Control vs. Flexibility

One prominent tension emerges around the shift to centralized planning. The introduction of the electric trucks requires significantly more planning as emphasized by the informants. Limited range, longer charging time, and insufficient charging availability in more strategic areas create operational constraints, which demand a higher level of control and optimization to ensure that the operations of the electric truck remain feasible and efficient. Consequently, decision-making starts to shift to be more centralized supported by digital route optimization tools.

However, the increased control challenges the established practices by the industry where the detailed planning and optimization typically relies on drivers level. Several informants underlined the drivers' strong preference for maintaining independence in planning and managing their workload which correlates with drivers satisfaction. The shift towards centralized, digitally assisted planning may therefore create internal resistance and require adjustments in organizational structures and responsibilities.

To address these tensions, one potential solution emerged throughout the project where electric trucks are prioritized for short-distance routes where their limitations are less critical and planning remains relatively simple. Many informants highlighted the significantly increased competitive performance of the electric truck for such transportation. This resembles the literature of Geels

(2002) where a niche market evolves for the new technology. Furthermore, in combination with the traditional diesel trucks, which in turn work well for longer hauls, it could create a state of hybridization where operations of the new and incumbent technology coexist (Geels, 2002).

However, this solution also introduces new tensions with diesel truck operators, who may lose access to short-haul filler loads that support their operational efficiency. In this case the forest company may play a crucial role. By providing short-distance routes and adapting the pricing models, incentives can be created that allow transport companies to naturally optimize their operations by vehicle allocation to choose the electric trucks for the short-distance routes over its diesel counterpart.

6.3.2 Tension 2: Project vs. Business

Another important tension concerns the role of the TREE project itself. Among the actors, there are varying perceptions of whether the project serves primarily as a temporary pilot or learning platform, while others see it as an essential step toward long-term business adoption and are already experimenting with potential business models outside the formal project boundaries.

Related to this, differences in transparency were observed. Inside the project, collaboration and information sharing seem to be open, however, some actors remain cautious about continuing such collaboration efforts post the project period. This leads to differences in levels of commitment, ambition, and openness in information sharing.

Several informants also pointed out that the project sometimes lacks a real-world perspective, emphasizing the need to start addressing business model questions and prepare for actual market conditions if the transition is to scale successfully. This reflects the broader challenge of moving from experimentation to market readiness in transitions, where ecosystem actors may need to align their efforts both strategically and operationally to enable scalable business models.

6.3.3 Paradox 1: Competition vs. Collaboration

Results show that competition between the actors is present and seen as a barrier to accelerate the transition. At the same time, literature and some informants both highlight that collaboration is key to the transition. The second aggregated dimension derived from the data also pointed at this issue,

as the gap between the experimented project and the envisioned resulting business is seen as too big in some cases. The collaboration taking place now cannot be transferred to or be part of the resulting business model, because of the competition between the actors, which is shown in Tension 2 between project and business. Implementing optimization software, which was considered by many as crucial for the acceleration of the transition, is also to be hindered by competition according to some informants. Ecosystems are collaborative by definition, but also competitive within. While it is posed as a potential tension between actors, it is not mentioned as a paradox (Adner, 2016). While it is quite clear in this case that competition hinders collaboration, it might be slightly generic as the terms are very broad. Most likely, there are specific aspects of collaboration and competition that are paradoxical.

Van Bommel (2018) presents some paradoxical tension, where one is considering short-term and long-term, and another considers sustainability- and business logic. Both of these seem applicable to this situation, as the long-term focus in this case aligns with the sustainability focus, and short-term perspective aligns with the business logic. Informants highlighted that the purpose of the project was to gain advantage and find cost-effective solutions for when the transition really happens, indicating a long term perspective. The collaboration that the sustainability logic brings then, or needs, is not compatible with the current business logic.

6.3.4 Paradox 2: Dependency vs. Change

Another paradox relates to where the responsibilities for the transition is perceived to lie. Most informants agree that transport leaders and drivers are the ones most directly affected by the transition, which was indicated by the emerging internal shift in the transport companies due the complexity of the new technology and the resulting changing operational demands. However, there was also a recurring theme that the forest companies should be the ones to take the lead, bear the risks, and provide the necessary conditions for change.

This creates a paradoxical situation where the point of change is perceived to be the forest company, while the actual need for adoption lies in the transport companies. In this way, those who are most affected by the change are seen as dependent on those less directly impacted by it to act as the driver of change. This resonates somewhat with literature of Dahlgren and Ammenberg (2022) again, where small haulers are more innovative. If the transport companies embrace the fact that they need to change, it might reduce the dependency

on the forest companies. However, if that shift in mindset does not occur, there might be a greater pressure on the forest company to continue creating incentives.

6.4 Reflections and Limitations

The *theoretical framework* built around transition theory, business model development and innovation ecosystems, offered useful lenses for analyzing the challenges in the sustainable transition to electrified forest transportation. These perspectives helped to understand both firm-level dynamics and broader ecosystem interactions. However, it is important to acknowledge that the selected frameworks may not capture all relevant factors influencing the transition. Other perspectives, such as institutional theory, socio-cultural change, or technology diffusion could have provided a more holistic insight in the analysis.

From a *methodological standpoint*, the study followed a qualitative, case-based approach, using semi-structured interviews as the source of main data. The Gioia methodology enabled a systematic and transparent coding process, which provides rigor by remaining close to the data from the interview transcripts while also identifying abstract patterns (Corley and Gioia, 2004). However, this approach is interpretive by nature, and findings may be shaped by our own analytical lens and prior knowledge. Furthermore, while the coding method is suggested to approach with an open mind, our prior literature study may have influenced how the authors identified and interpreted patterns, aligning more with Eisenhardt's (1989) logic of iterating between data and theory.

While the case is based on the TREE project, the *empirical setting* of the master's thesis focused only on a single ecosystem within a specific geographical location with special conditions. The specific site was relatively advanced in terms of available infrastructure which may limit transferability to other demonstration sites within the project. Additionally, in the context of Sweden, factors such as policies and governmental support can differ from other countries, which may further limit generalizability.

In terms of *data gathering*, while all focal actors of the ecosystem of the specific site were covered, some companies were only represented by only one particular person. Consequently, the data might not reflect the general view of the company on the project. Some differences have been observed depending on the position of the informants and how closely they are involved in the project. While CEOs were generally more positive towards the transition and

had a clear vision, others were perceived a bit more reluctant. The inclusion of external informants was targeted to mitigate this to some extent by including an outside perspective to test the findings. However, the sample remained limited due to the limited number of actors who were in the TREE ecosystem but still related to the ecosystem of the specific site. Therefore, a possibility to test the transferability of the findings could be to interview actors of an ecosystem from a different demonstration site within the TREE project.

Finally, it is important to note that this study was conducted during an *early phase of the transition*. Specifically, only one electric timber truck was up and running by the start of the master's thesis. Furthermore, due to the quite special weather conditions in Sweden, the road conditions change drastically from winter to summer. As a result, the challenges identified might only reflect the initial state of the transition at the time when the collection of main data has started, and therefore are likely to evolve over time as weather changes, business models stabilize, and actor relationships shift. To better cover the dynamic of the transition and mitigate the influence of early stages in a transition, it could have been beneficial to conduct interviews with the same actors at a later point of time. However, due to the limited timeframe of the master's thesis, this was not possible.

Despite these limitations, findings can be somewhat generalized when adapting the scope of a transition with similar boundaries, such as early-stage considerations and the occured tensions within an ecosystem consisting of actors competing in the same sector. Regardlessly, it is believed that the study offers meaningful contributions by capturing early signals, emerging tensions, and paradoxes that can inform both future research and practice in similar transition contexts.

6.5 Future research

Possible future research could include the possibilities of large-scale optimization of fleets and its implication on sustainable transitions in road transportation. Another recommendation is to study the impact of partnering with bigger and smaller transport providers, and the long-term effects that has for sustainable transitions. The tensions and paradoxes could also be studied more thoroughly and frameworks developed including strategies to handle these particular tensions.

Future research should focus on the importance of articulating the additional value created by this kind of transition. Firstly for the whole ecosystem, as it would serve as a foundation for further value distribution

within the ecosystem, to then study the impact and competitiveness of the individual organizations.

Chapter 7

Conclusion

This master's thesis set out to explore challenges of a sustainable transition of an industry and its implications on both the individual organization and the ecosystem. The study took an explorative path to try and uncover details of a very specific and unique case in the Swedish forest industry. Literature has emphasized the importance for the individual firm through business model development, the system-level perspective of transition theory and the interrelation between firms in the ecosystem. In order to describe the aspects of the most prominent challenges, the ambition was to combine these literature streams to uncover the considerations for both the individual organisation and the ecosystem as a unit, and their interrelation. The TREE project and the electric timber truck presented a case where such aspects promised to be identified.

The results gave five aggregated dimensions which revealed some of the challenges that the ecosystem is facing. The technological barriers were presented first, highlighting the bottlenecks that lower performance posed. Second, there was the issue of the different perceived purposes of the project for the different actors, indicating that alignment of activities and strategy would be needed. This was later discussed as a potential tension and paradox, where business and project, and collaboration and competition were seemingly contradicting each other. A third dimension concluded that the focal actor in this case is the driver of the transition and seen as the actor that needs to take risks. Changing decision-making and processes of transport companies was a fourth dimension, which together with the risk-sharing supported the tension between control and flexibility, and the paradox of dependency and change. The transport companies and haulers advocated for flexibility, but were in increasing need of control for the transition to happen, which requires

risk that they expect the focal actor to take. A fifth dimension highlighted the resistance to change within the ecosystem which was stronger among drivers, haulers and transport companies. These contribute to the research of sustainable transitions by highlighting potential tensions and paradoxes that can occur in similar settings, either in that of the roles of the actors, or the same kind of technological transition.

This master thesis projected and emphasized several aspects of a sustainable transition in an industry. The identified, most prominent challenges with developing business models were the implementation of a new business model and the changing competitive advantage of some firms. For the ecosystem, capturing the collective value and the coexistence of the new and the old technology stood out as the greatest challenges. This aided an emerging theme of digitalization and optimization, which seemed to be a requirement for this ecosystem to transition. In these challenges, the interesting relationship between transport buyer and transport provider was also present. Contrary to some previous literature, suppliers of a bigger size posed greater potential as partners in the transition for the transport buyer in this case.

These insights can serve as important considerations for other case studies in sustainable transitions by highlighting the challenges that arise in this stage of the transition, and in a case of this setup. It is important to establish what the collective value is, how to capture it, and who should take risk. There is likely going to be resistance and tensions, even within organizations. Even paradoxes can be faced, which need to be attended if the transition is to continue. Ecosystems facing similar transitions should consider that overarching changes and implementations to align the ecosystem might be crucial for the transition, such as digitalization.

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Appendix A

Interview guide

Interview Guide

Introduction:
- What is your position in the company?
- What does the structure look like in the company?
- What are you responsible for in the project?

Project related:
- What is the status of the project in site 4?
- Who are your main partners and what do your relationships with them look like?
- What do the operations look like on a daily basis? What are the different steps from planning to finishing a job?

Overview barriers
- What are the most pressing barriers and who is affected by it?
- How are you trying to solve these?

Electric trucks
- What are the main differences between the new electric trucks and diesel trucks and how do they affect you and other stakeholders?
- How are these uncertainties reduced?
- What does the introduction process look like for an electric truck from planning to start of operation?

Infrastructure:
- What does the current charging situation look like? What is lacking and what is working well?
- How are you planning to set up the next charging stations? What factors do you have to consider and who will be part of it?

Actors and collaborations
- Does everyone have the same idea about going forward with the development of Site 4?
- Do you look at new actors to be involved?
- A system challenge seems to be lack of flexibility. What's needed to make it more flexible? Who needs to change?

Business models:
- How do you work with business model development?
- How do you handle a new business model alongside normal operations?
- Is there a demand for sustainable transport?
- How can you capitalize on being a pioneer?

Ending

- In what scenario would this system be beneficial?
- What are the next steps in the project? What outcome are you hoping for?
- Thank you for your time and contribution! We will get back to you with the results
- Any others to interview?

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